

7. SIGNAL RECONSTRUCTION AND SPECTRAL ANALYSIS

Experiment 7.1

Reconstruction of Signal from Samples (Ideal Interpolation) .

Aim :

Reconstruct a bandlimited signal using sinc interpolation.

Apparatus Required:

PC with MATLAB.

Algorithm:

1. Sample bandlimited signal.
2. Apply sinc interpolation formula.
3. Compare with original.

Program:

```
% Reconstruction of Signal from Samples
```

```
clc;
```

```
clear;
```

```
close all;
```

```
% Original signal frequency
```

```
fm = 5; % Hz
```

```
% Sampling frequency
```

```
fs = 25; % Hz
```

```
% Continuous-time signal
```

```
t = 0:0.001:1;
```

```
x = sin(2*pi*fm*t);
```

```
% Sampling
```

```
ts = 0:1/fs:1;
```

```
xs = sin(2*pi*fm*ts);
```

```
% Reconstruction using interpolation
```

```
xr = interp1(ts, xs, t, 'spline');
```

```
% Plotting
```

```
figure;
```

```
subplot(3,1,1);
```

```
plot(t, x, 'LineWidth', 2);
```

```
grid on;
```

```
title('Original Signal');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

```
subplot(3,1,2);
```

```
stem(ts, xs, 'filled');
```

```
grid on;
```

```
title('Sampled Signal');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

```
subplot(3,1,3);
```

```
plot(t, xr, 'LineWidth', 2);
```

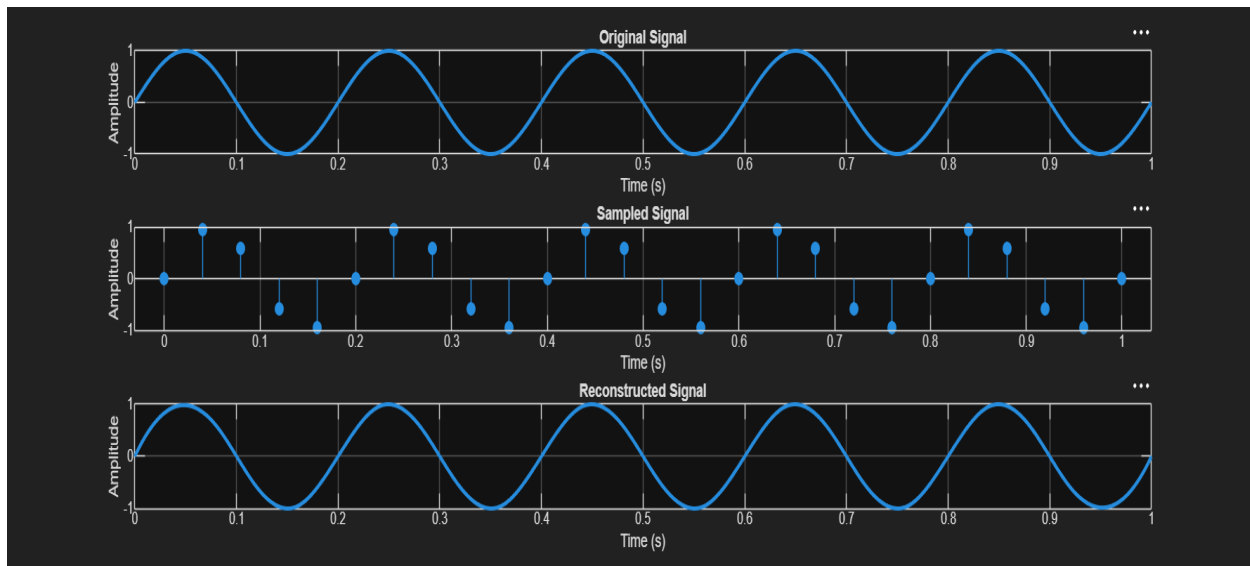
```
grid on;
```

```
title('Reconstructed Signal');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

Output waveform:



Result:

The sinusoidal signal was sampled at **25 Hz**, which is greater than the Nyquist rate of **10 Hz**. Using interpolation, the original signal was reconstructed successfully, demonstrating accurate signal recovery from its samples in accordance with the Sampling Theorem.

Experiment 7.2

Fourier Series of Half-Wave Rectifier Output .

Aim:

To compute and analyze the Fourier series representation of the output of a half-wave rectifier. The Aim is to understand how the Fourier series can decompose a non-sinusoidal periodic signal into its constituent sinusoidal components (harmonics), and to observe how these harmonics contribute to the overall signal.

Program:

```
% Fourier Series of Half-Wave Rectifier Output
```

```
clc;
```

```
clear;
```

```
close all;
```

```
% Time axis
```

```
t = 0:0.001:2*pi;
```

```

% Half-wave rectified signal

x = sin(t);

x(x < 0) = 0;

% Fourier Series approximation

N = 10; % Number of harmonics

a0 = (1/pi) * trapz(t, x);

fs_signal = a0/2;

for n = 1:N

    an = (1/pi) * trapz(t, x .* cos(n*t));

    bn = (1/pi) * trapz(t, x .* sin(n*t));

    fs_signal = fs_signal + an*cos(n*t) + bn*sin(n*t);

end

% Plotting

figure

subplot(2,1,1);

plot(t, x, 'LineWidth', 2);

grid on;

title('Half-Wave Rectified Signal');

xlabel('Time');

ylabel('Amplitude');

subplot(2,1,2);

plot(t, fs_signal, 'LineWidth', 2);

grid on;

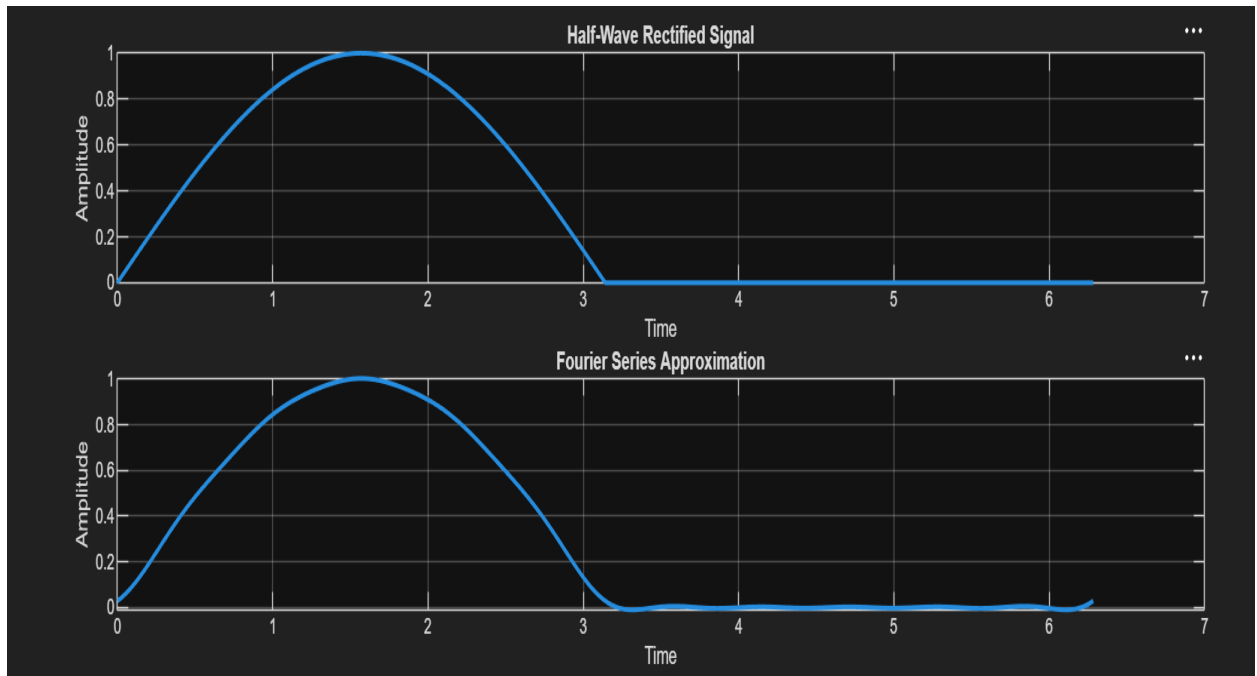
title('Fourier Series Approximation');

xlabel('Time');

```

```
ylabel('Amplitude');
```

Output waveform:



Result:

The Fourier Series of the half-wave rectified signal was obtained using MATLAB. The reconstructed waveform using Fourier coefficients closely matched the original half-wave rectified signal, demonstrating the Fourier Series representation of periodic signals.